

Patterns in Habitat-Specific Estuarine Primary Productivity Fail to Predict Distinctions at two Higher Trophic Levels

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Human threats to estuaries lead to preferential protection and restoration of specific habitats, especially marsh and seagrass. Habitat-by-habitat syntheses of all available data from the eastern U.S. reveal that, although these two habitats exceed all others in primary production of foods suitable for consumption by herbivores and detritivores by factors of 2-9, total net production over the lowest three trophic levels is 4-10 times higher in oyster reefs than in any other estuarine shoreline habitat. Salt marsh and seagrass habitats exceed tidal flats in total net productivity by only 1.6-2.5 times, using empirically inferred ecological efficiencies of 26-27%, or by undetectable amounts when higher trophic levels are weighted more by assuming Lindemann efficiencies of 10%. High cross-trophic-level interconnectivity and benthic microalgal production help explain the surprising failure of vegetated habitats to dominate production at higher trophic levels. Estuarine habitat conservation demands a more holistic approach to sustain food web services, including treating oysters as habitat rather than as an exploitable commodity.

Much ecological conservation targets particular valued habitats for protection against threats and for restoration (1, 2). Estuaries and their constituent habitats experience especially high levels of threat from numerous human-driven stressors (3, 4), including overfishing, coastal development, eutrophication, chemical contamination, and growing challenges to coastal wetlands arising from sea level rise (5). Among the many ecosystem services provided by estuaries (6), high biological production in food chains of value is widely recognized as a function worthy of management support. The history of habitat conservation and restoration within temperate estuaries demonstrates triage favoring salt marsh and seagrass beds, each of which displays visual evidence of past primary productivity in stands of vascular plant vegetation. High rates of primary productivity and provision of above- and sometimes also below-ground structure historically have been used to justify the conservation focus on these two estuarine habitats (7, 8).

Valuation of habitats within the estuary, and by extension in other ecosystems, on the basis of primary production and standing plant structure suffers from four fundamental limitations. First, vascular plants represent only the largest of primary producers and their distributions do not necessarily match among-habitat patterns in production of more cryptic plants, the benthic microalgae, which are directly and more efficiently consumed by secondary producers (9, 10). Second, the selection of the primary producer level at which to evaluate relative habitat importance is largely dictated by convention. Fixation of carbon in vascular plants is assumed to fuel high production of the entire food web, but much of this carbon is recalcitrant (11) and contrasts of production at higher trophic levels may reveal different patterns among habitats. Third, structural habitat complexity is provided not only by emergent plants but also by reefs of bottom invertebrates, especially oysters and polychaetes. Fourth, while estuarine bottom can be reasonably partitioned into separate habitats, connectivity among habitats is high (12), resulting both from passive transport of materials forced by astronomical tides, winds, and freshwater run-off from land and also from active movements among habitats by mobile animals. Consequently, the estuary might best be treated as a system of interconnected habitats, and relative habitat value evaluated by a metric that includes multiple trophic levels to capture processes that reflect production of food in some habitats and its consumption in others.

We report results of syntheses and analyses of all accessible data from estuaries of a common biogeographic region, the eastern seaboard of the U.S. from Laguna Madre, Texas to Cape Cod, Massachusetts, on net production in each of the bottom three trophic levels in each estuarine shoreline habitat. Habitats considered are, ordered broadly by elevation, *Spartina* salt marsh, intertidal flat, oyster reef, *Zostera* (and *Thalassia*) seagrass beds, and shallow subtidal flat. Numbers of replicate data sets in the syntheses totaled the following by habitat: for primary production 45, 3, 1, 14 (16), and 2; for secondary production 8, 11, 9, 10 (3), and 43; for tertiary production 16, 3, 31, 4 (13), and 17. We use these syntheses to construct ratios of 1⁰:2⁰:3⁰ production within each habitat and compare them to energetic expectations based on Eltonian pyramids (13), first assuming Lindemann (14) ecological efficiencies of 10% and second inferring ecological efficiencies from our observed average ratios of production between adjacent trophic

levels, to reveal patterns and magnitudes of connectivity among estuarine shoreline habitats. We then convert production of each trophic level to equivalent units of primary production of the suitable foods for herbivore or detritivore consumption required to have produced the observed biomass of that higher trophic level. Finally, we average these ratios across the three trophic levels and also construct ratios of sums of the primary production equivalents across all three trophic levels for each habitat to serve as synthetic production metrics with which to compare the biological production values of each estuarine habitat.

Production is perhaps the most fundamental ecological process and therefore preferred over static structural measures like abundance or biomass as one metric of ecological function (14). As a process, production of entire trophic levels is never directly measured but instead approximated by one of several techniques. Although each approach possesses uncertainty over the accuracy of the absolute value of production, by standardizing estimation methods across all habitats, the comparisons between habitats are robust to systematic errors associated with estimation method (details of methods and magnitudes of variances given in SOM). Because the value of primary production lies in its support of food chains, we chose to measure primary production as the estimate of the net annual biomass (in g ash-free dry mass $\text{m}^{-2} \text{y}^{-1}$) of foods produced in forms that can be assimilated by herbivores and detritivores. This choice is consistent with data needs for computation of ecological efficiency of transfer of production between trophic levels. Productivity was estimated for each of five benthic (bottom) habitats of estuarine shorelines. Benthic microalgae are directly consumed by herbivorous invertebrates and thus all of their production was included. Vascular plants in the eastern seaboard estuaries, however, mostly enter detrital food chains, in which plant detritus is first transformed into usable foods for detritivores by conversion to fungal and bacterial decomposers, at the expense of inefficiencies from energetic losses during conversion (10, 11). Consequently, we account for this necessary conversion into microbes in computing the production of foods from vascular plants that are assimilated by detritivores, discounting the fraction of vascular plant production that fuels detrital food chains by the ecological conversion efficiency from salt marsh grass or seagrass into edible microbes.

We estimated 2^0 and 3^0 production (P) in each habitat by multiplying data on biomass (B) by P:B ratios taken from the literature. For 2^0 producers, macrobenthic biomass (B) of all herbivores and detritivores was multiplied by published phylum-specific P:B ratios (e.g., 15, 16): 4.8 for Annelida; 5.7 for Arthropoda; 11 for Chordata; 5.3 for Cnidaria; 0.3 for Echinodermata; 2.9 for Mollusca; 4.3 for Nemertea; and 4.0 for Porifera. In *Spartina* marshes, we included terrestrial 2^0 production by insects like grasshoppers and in seagrass beds we included sea urchin and fish herbivory. Although a previous synthesis of 2^0 marine benthic production (16) suggests no latitudinal difference in P:B ratios over the modest geographic range covered by our data sets, we also made geographic contrasts of net 2^0 production of benthic sedimentary invertebrates in our data (SOM Fig. S6), revealing only some modest potential of confounding by latitude. Finally, we synthesized data on biomass of nektonic 3^0 producers from the five estuarine

habitats and converted fish biomass to production by multiplying by a P:B ratio of 2 and crustacean biomass by 3 (11).

Contrasts of productivity across estuarine habitats revealed that differences among habitats depend strongly on trophic level (Table 1). The two habitats defined by emergent stands of vascular plants, the *Spartina* marsh and *Zostera* or *Thalassia* seagrass beds, possess high rates of net annual 1^0 productivity, sufficient, even after discounting production of vascular plant detritus by conversion to microbes suitable for detritivore assimilation, to elevate them well above tidal flats and oyster reefs by factors of 2-9. Nevertheless, benthic microalgal production on intertidal and shallow subtidal flats also contributes substantially to food web support (17, Table 1). Such microalgae are consumed directly by herbivorous invertebrates without passing through a microbial intermediate and thus transfer a higher fraction of the carbon they fix to higher trophic levels (Fig. S1, Table S2). The inclusion of benthic microalgae, which are quite productive on tidal flats, reduces the magnitude of dominance by the marsh and seagrass. Microalgae and some macroalgae growing on oyster shells and on sediments interspersed among oysters on reefs likewise make substantial contributions to support food webs (Table 1). In contrast, 2^0 productivity is dramatically higher on oyster reef habitat than all others by about 200-fold, driven by oysters and other benthic macroinvertebrates associated with the stable hard substratum (18). Differences in 2^0 productivity among the other four estuarine habitats are undetectable. Similarly, average annual 3^0 productivity varied little among estuarine habitats, with oyster reef higher by a factor of 2 (18) and *Thalassia* seagrass beds perhaps lower than the others (Table 1).

Expressing the productivity values across trophic levels as ratios, first assigning 100 to the 1^0 production of each habitat, provides insight both into connectivity among habitats as reflected in departures from the 100:10:1 ratios expected on the basis of the Eltonian pyramid of production (13) and also into the accuracy of assuming these Lindemann (14) ecological efficiencies of 10%. The oyster reef habitat demonstrated the greatest deviation from expectation, with 2^0 productivity 5 times as high as 1^0 , not the one tenth expected (Table 1). This deviation reflects the massive subsidy of primary production into this habitat as phytoplankton, imported by tidal currents, forms the bulk of the oyster diet (18). The only habitat in which the observed production ratios closely matched all Lindemann expectations is seagrass. Across the salt marsh and both tidal flat habitats, 3^0 productivity was consistently higher than expected on the basis of net annual 2^0 productivity by a factor of 3-4 (Table 1). In intertidal habitats (salt marsh, intertidal flat, and about half of the oyster reefs sampled), the 3^0 production estimate may be biased upwards by the assumption that the nektonic fishes and crustaceans necessarily sampled at high tide conducted almost all their feeding during high-tide occupation of intertidal habitats. To the degree that this assumption is violated, the anomalously high 3^0 production of intertidal habitats may reflect interconnectivity among habitats achieved by active migration of predatory fish and crustaceans and passive transport of drift algal subsidies with the tides. Nevertheless, departure from this assumption fails to explain the similarly elevated $3^0:2^0$ productivity ratio over subtidal flats, suggesting a possibly greater efficiency of transfer from benthic herbivores/detritivores to their predators than the Lindemann 10%. Ecological efficiency depends on the fraction of production that is

consumed by the next higher trophic level as well as the energetic balance between food assimilation and respiration. A synthesis of predator exclusion experiments in estuaries demonstrated that intense predation by nektonic crustaceans and fishes controls abundances of benthic invertebrates in tidal flat habitats but not in seagrass (20), implying that the secondary producers of tidal flats, soft-bottom benthic invertebrates, are unusually completely consumed and thus their production energy is more effectively transferred up the food chain.

The two tidal flat habitats also exhibited greater 2^0 productivity than expected based on applying the 10% Lindemann efficiencies to 1^0 productivity by factors of 3-6 (Table 1). This departure in part represents quantification of tidal importation of 1^0 production subsidies from three sources: phytoplankton in the water column; vascular plant detritus from tidal marsh and seagrass habitats; and drift algae. The 100:9.6 ratio of 1^0 production to 2^0 production in *Spartina* marsh, however, failed to demonstrate the large excess 1^0 production expected if this habitat were exporting vascular plant detritus to enhance 2^0 productivity of tidal flats. Higher benthic bivalve 2^0 production on shallow subtidal flats with than on those without *Spartina* marshes above them demonstrates the importance of this path of connectivity (12). Our inability to confirm substantial excess 1^0 production relative to what is required to produce observed levels of 2^0 producers in *Spartina* marshes may reflect geographic and other spatial differences in our sources of 1^0 and 2^0 production data. None of the individual marsh studies also measured production at multiple trophic levels, so data used in our synthesis of 2^0 productivity could by accident or undetected bias have been derived from intrinsically more productive sites than data for 1^0 productivity. Data and analyses in the SOM reveal that the geographic differences between sources of 1^0 and 2^0 data contribute only a small possible bias. Consequently, the absence of a large excess in 1^0 productivity relative to 2^0 productivity in the salt marsh and seagrass bed habitats despite widespread observation of detritus export to other unvegetated estuarine habitats implies that using Lindemann ecological efficiencies of 10% understates the true transfer efficiencies in an estuarine shoreline ecosystem.

Choice of appropriate ecological efficiencies of transfer between trophic levels greatly affects the computation of total trophic-level adjusted productivity over all three lowest trophic levels, a reasonable synthetic basis for quantitative habitat comparisons. Conversion to comparable units of primary production required to achieve the demonstrated production of each trophic level is the first step in this computation process. We made two separate sets of calculations, one assuming the traditional Lindemann (14) ratios of 10% and the other using empirically inferred ratios from our syntheses. Previous reviews of animal production and respiration suggest that carnivorous invertebrates exhibit ecological efficiencies up to 21-56%, while ecological efficiencies of endotherms (birds and mammals) are low, around 1-3%, because of high respiration required for body warming (21), and that those of vertebrate ectotherms (fish) are around 10% (22, 23). Non-insect invertebrates display a mean ecological efficiency of 25%, varying by trophic group, with herbivores at 21%, carnivores at 28%, and detritivores the most efficient at 36% (23). Averaged across all habitats in our synthesis except oyster reefs, 3^0 production represents 27% rather than 10% of 2^0 , and 2^0 production 26% of 1^0

production (Table 1). We eliminated the oyster reef data from these computations because the oysters that dominate the 2⁰ production in this habitat feed largely on subsidies of phytoplankton from outside the habitat, and likewise because little of the diet of the 3⁰ producers in this habitat is comprised of oysters. Using the observed 27% 2⁰ to 3⁰ and 26% 1⁰ to 2⁰ ecological conversion efficiencies in summing production equivalents across the three trophic levels has the effect of reducing the weighting of 2⁰ and 3⁰ production relative to 1⁰ because less fixed carbon is required to produce each of the next two higher trophic levels. These empirically derived ecological efficiencies seem to represent a more justifiable choice than Lindemann efficiencies because they are supported by our extensive data synthesis and by similarly high empirically inferred mean efficiencies for invertebrate herbivores/detritivores and carnivores in the previously published syntheses. Alternatively, greater weighting of higher trophic levels may be intentionally applied by environmental managers to express an explicitly enhanced valuation of the higher trophic levels.

Between-habitat ratios of productivity incorporating all three trophic levels vary with metric chosen, although all but one ranking of pairs of habitats are preserved independent of metric (Table 2). Averaging ratios across the three trophic levels avoids need to assign energetic efficiencies of trophic transfer but involves statistical discomfort of averaging ratios. For every metric, the relatively high net productivity of oyster reef habitat was evident, 8-9 times (using averages), 7-8 times (using Lindemann efficiencies) and 4-10 times (using inferred efficiencies) other estuarine shoreline habitats (Table 2). These results provide unambiguous quantitative evidence that oyster reef habitat delivers greater food web services per unit area than any other estuarine shoreline habitat. *Spartina* marsh exhibited both higher average productivity ratios and higher total production using inferred transfer efficiencies than intertidal and shallow subtidal flats by factors of 1.6-1.7, as compared to only 1.2 assuming Lindemann efficiencies (Table 2). Seagrass beds also were revealed to be more productive than both tidal flat habitats by average productivity ratios of 2-4 and ratios of total production of 1.8-2.5 using inferred efficiencies, as compared to virtually identical total productivities applying Lindemann efficiencies (Table 2). Ratios of *Spartina* marsh to seagrass habitat production averaged across all three trophic levels were slightly greater than unity at 1-1.7, while the corresponding ratios of total production under Lindemann efficiencies were 1.2-1.3. Using inferred ecological efficiencies, however, the rank order of these two habitats was reversed, with seagrass habitat emerging as more productive than *Spartina* marsh by a factor of 1.2-1.4, implying ambiguity in reaching definitive conclusions about how these two vegetated estuarine habitats differ from one another in total productivity across three trophic levels.

Adding syntheses of productivity at higher trophic levels to what has been already widely reported for primary production of vascular plants provides important insights into the interconnected nature of estuarine habitats. At both the second and third trophic levels, vegetated habitats fail to exhibit the dominance over intertidal and shallow subtidal sand flats that characterizes their primary production (Table 2). Because estuarine primary production is valued in large part for its support of higher trophic levels (6), the apparent redistribution of much of this marsh and seagrass production into tidal

flat habitats for its conversion to higher trophic levels of value emphasizes need to protect not only habitats where carbon is fixed but also those where the fixed product is consumed. We propose that high passive and active connectivity among estuarine habitats leads to the unexpected virtual equivalence of all habitats except oyster reefs in production of the animals at the second and third trophic levels, providing new justification for ecosystem-based management of estuaries as cohesive systems of interconnected habitats. Estuaries are also connected to upland, riverine, and marine ecosystems through passive transport of inorganic and organic materials and through active movements of animals like most fishes and birds, so discrepancies among trophic levels in between-habitat productivity relationships within estuaries may also reflect across-ecosystem subsidies and losses (24).

Oyster reef habitat has been dramatically depleted and degraded (25, 26) with important loss of its ecosystem services of sustaining water quality by intense filtration that can suppress eutrophication (3), of provision of biogenic habitat structure for fishes (27), and of sustaining high biodiversity and large economic values of nature's services (28, 29). The trophic level at which oyster reefs achieve their extraordinarily high productivity relative to other habitats is the herbivore level (Table 2). This high production is driven by oysters and other marine invertebrates, which are facilitated by the provision of stable, emergent hard substratum, analogous to the generally far higher biomass and production of marine intertidal invertebrates per unit area on rocky shores than on ocean beaches and in other sedimentary environments (30). Our synthetic review of the productivity of estuarine habitats supports expansion of efforts to protect and restore oysters for their habitat value instead of exploiting them as a commodity and to enhance conservation of tidal flats in recognition of their importance in the transfer of high primary production of the entire estuarine shoreline system to animals at the next two higher trophic levels.

References and Notes

1. W.V. Reid, K.R. Miller, *Keeping Options Alive: the Scientific Basis for Conserving Biodiversity* (World Resources Institute, Washington, D.C 1989).
2. National Research Council, *Understanding Marine Biodiversity* (National Academy Press, Washington, D.C. 1995).
3. J.B.C. Jackson *et al.*, *Science* **293**, 629 (2001).
4. H.K. Lotze *et al.*, *Science* **312**, 1806 (2006).
5. J.G. Titus *et al.*, *Coast. Manag.* **172**, 189 (1991).
6. Millennium Ecosystem Assessment, *Ecosystems and Human Well-Being: Wetlands and Water* (World Resources Institute, Washington, D.C. 2005).
7. E.P. Odum, *NY State Conserv.* **29**, 60 (1961).
8. J.M. Teal, *Ecology* **43**, 614 (1962).
9. S.Y. Newell, D. Porter, in *Concepts and Controversies in Tidal Marsh Ecology*, M.P. Weinstein, D.A. Kreeger, Eds., (Kluwer Academic Publ., Boston, 2000), pp. 159-185.
10. R.T. Kneib, *Mar. Ecol. Prog. Ser.* **264**, 279 (2003).

11. J. Cebrian, *Am. Nat.* **154**, 449 (1999).
12. R.D. Seitz *et al.*, *Mar. Ecol. Prog. Ser.* **326**, 11 (2006).
13. C.S. Elton, *Animal Ecology* (Sidgwick and Jackson, London, 1927).
14. R.L. Lindemann, *Ecology* **23**, 399 (1942).
15. R.J. Diaz, L.C. Schaffner, in *Perspectives on Chesapeake Bay, 1990: Advances in Estuarine Sciences*, M. Haire, E.C. Krone, Eds., (Ches. Bay Res. Consort., Gloucester Point, VA, 1990), pp.25-56.
16. M. Cusson, E. Bourget, *Mar. Ecol. Prog. Ser.* **297**, 1 (2005).
17. M.J. Sullivan, C.A. Currin, in *Concepts and Controversies in Tidal Marsh Ecology*, M.P. Weinstein, D.A. Kreeger, Eds., (Kluwer Academic Publ., Boston, 2000), pp. 81-106.
18. Meta-analytic one-way ANOVAs on 2⁰ production data demonstrated significance of differences among habitats ($F_{5,78}=30.7$; $p<0.0001$) and on 3⁰ production data significance of differences among habitats ($F_{5,78}=5.5$; $p=0.0002$). Transformations failed to homogenize variances for 2⁰ production data, but the extremely low p-values nonetheless support the inference of significance. The 3⁰ production data required no transformation. Post hoc SNK contrasts show that in 2⁰ and 3⁰ production oyster reefs segregate significantly ($p<0.05$) from all other habitats, which cannot be distinguished.
19. E.B. Haines, in *Ecological Processes in Coastal and Marine Systems*, R.J. Livingston, Ed., (Plenum Press, New York, 1979), pp. 35-46.
20. C.H. Peterson, in *Ecological Processes in Coastal and Marine Systems*, R.J. Livingston, Ed., (Plenum Press, New York, 1979), pp. 223-264.
21. S.L. Pimm, *Food Webs* (Univ. Chicago Press, 1982, 2002).
22. S. McNeil, J.H. Lawton, *Nature* **225**, 472 (1970).
23. W.F. Humpfreys, *J. Anim. Ecol.* **48**, 427 (1979).
24. G.A. Polis, W.B. Anderson, R.D. Holt, *Annu. Rev. Ecol. Syst.* **28**, 289 (1997).
25. B.J. Rothschild *et al.*, *Mar. Ecol. Prog. Ser.* **111**, 29 (1994).
26. H.S. Lenihan, C.H. Peterson, *Ecol. Appl.* **8**, 128 (1998).
27. H.S. Lenihan *et al.*, *Ecol. Appl.* **11**, 764 (2001).
28. C.H. Peterson, R.N. Lipcius, *Mar. Ecol. Prog. Ser.* **264**, 297 (2003).
29. J.H. Grabowski, C.H. Peterson, in *Ecosystem Engineers: Plants to Protists*, K. Elsevier, Amsterdam, 2007), pp. 281-298.
30. C.H. Peterson, *Am. Sci.* **79**, 236 (1991).
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Table 1. Average (± 1 SE) productivity (g AFDM m⁻²y⁻¹) at the three lowest trophic levels in each of five estuarine shoreline habitats and within habitat ratios of 1⁰:2⁰:3⁰ production, derived from synthesis of available data from Texas to Massachusetts, USA. Primary productivity was adjusted to account for necessary microbial transformation of vascular plant detritus to produce suitable foods for detritivores (methods in SOM). Weighted sums across the three trophic levels are computed in units of primary production required to produce observed average productions, assuming first Lindemann 10% ecological efficiencies at each conversion to the next highest trophic level and second empirical efficiencies derived from averaging within-habitat productivities across all habitats except oyster reefs. No SE is provided for primary production because the value combines empirical data and modeled estimates.

Trophic level	Habitat				
	<u>Spartina marsh</u>	<u>Intertidal flat</u>	<u>Subtidal flat</u>	<u>Seagrass bed¹</u>	<u>Oyster reef</u>
Primary - 1 ⁰	688	222	135	960 1431	330
Secondary - 2 ⁰	66(± 6)	75(± 18)	73(± 17)	79(± 26) 89(± 14)	1635(± 360)
Tertiary - 3 ⁰	24(± 6)	22(± 12)	23(± 8)	15(± 4) 6(± 2)	47(± 5)
Ratio of 1 ⁰ :2 ⁰ :3 ⁰	100 : 9.6 : 3.5	100 : 34 : 10	100 : 54 : 17	100 : 7 : 0.9	100 : 505 : 14
Weighted sum by Lindemann 10% efficiency	3,748	3,172	3,165	3,250 2,921	21,380
Weighted sum by inferred efficiencies of 26 and 27%	1,284	824	743	1,483 1,859	7,288

¹- *Zostera marina*, then *Thalassia testudinum* beds. Ratio entry is averaged across both seagrass species.

Table 2. Ratios of productivity comparing pairs of estuarine shoreline habitats using average production data in Table 1.

Habitat Comparison	Trophic level			Average across 3 trophic levels	Based on 10% weighted sum	Based on 26-27% weighted sum
	1 ^o	2 ^o	3 ^o			
(1)Oyster reef to:						
Intertidal flat	1.5	21.8	2.1	8.5	8.4	8.8
Subtidal flat	2.4	22.4	2.0	8.9	8.5	9.8
<i>Spartina</i> marsh	0.5	24.8	2.0	9.1	7.0	5.7
Seagrass bed ¹	0.3, 0.2	20.7, 18.4	3.1, 7.8	8.0, 8.8	7.6, 7.3	4.9, 3.9
(2) <i>Spartina</i> marsh to:						
Intertidal flat	3.1	0.9	1.1	1.7	1.2	1.6
Subtidal flat	5.1	0.9	1.0	2.3	1.2	1.7
Seagrass bed ¹	0.7, 0.5	0.8, 0.7	1.6, 4.0	1.0, 1.7	1.2, 1.3	0.9, 0.7
(3) Seagrass bed ¹ to:						
Intertidal flat	4.3, 6.5	1.1, 1.2	0.7, 0.3	2.0, 2.7	1.0,,0.9	1.8, 2.3
Subtidal flat	7.1, 10.6	1.1, 1.2	0.7, 0.3	3.0, 4.0	1.0, 0.9	2.0, 2.5

¹ – Computations presented separately for *Zostera marina* and *Thalassia testudinum* respectively.

² – Weighting was done by energetic equivalents, using Lindemann 10% ecological efficiencies to convert production at each trophic level to the 1° production of usable foods required to produce that 2° or 3° production.

³ - Weighting was done by energetic equivalents, using inferred ecological efficiencies of 26%-27% to convert production at 1° to 2° and 2° to 3° levels respectively.